

Supplementary Information

Distilling solvation physics into AI for simulation-free hydration free energies

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Supplementary overview

This document reports the complete validation record supporting SolvAI. It separates the deployable structure-only model from candidate calculations that require simulation at inference, and it preserves non-improving physical-supervision experiments without making them the main narrative. All numerical values originate from `results/paper_metrics.json`; the corresponding molecule-level predictions are distributed with the repository.

Supplementary Table 1: complete primary comparison

Method	Simulation at inference	Random OOF	Family holdout	Scaffold holdout
Classical ARROW	yes	0.785	–	–
ARROW/PIMD8	yes	0.205	–	–
Previous structure-only	no	0.239	–	–
Narrow response	no	0.214	–	–
+ SMD(water)	no	0.202	–	–
SolvAI: + ConfSolv response	no	0.197	0.240	0.241
Nested feature selection	no	0.199	–	–

Table 1: MAE in kcal mol⁻¹. A dash denotes a comparison that was not defined for that fixed artifact. Only rows marked “no” satisfy the deployment requirement.

Supplementary Table 2: independent split repeats

Procedure	Split 1	Split 2	Split 3	Split 4	Split 5	Mean	SD
Fixed SolvAI	0.2074	0.1979	0.2085	0.2060	0.1991	0.2038	0.0049
Nested selection	0.2057	0.2067	0.2113	0.2076	0.2117	0.2086	0.0027

The five split seeds were 314159, 271828, 161803, 141421 and 173205. Each row is a complete five-fold OOF evaluation. No split was removed or replaced on the basis of its result.

Supplementary Table 3: master experiment table

Experiment	Scientific hypothesis	Data or response source	Inference	Leakage	OOF MAE	Interpretation	Retained
Previous structure-only	Molecular fingerprints and descriptors directly encode hydration	public hydration + RD-Kit/Morgan	SMILES	strict	0.23861	Establishes the pre-distillation baseline	yes, as base
Narrow response	Compact physical axes regularize the endpoint	CombiSolv-QM, Abraham, OpenFF, GBn2	SMILES	strict	0.21362	Broad but targeted response supervision transfers	yes
SMD water response	A water-specific continuum target supplies missing solvent response	MolSolv SMD(water)	SMILES	strict	0.20161	Largest single aligned gain	yes
ConfSolv response	Conformer reweighting and response distributions add complementary information	ConfSolv H ₂ O	SMILES	strict	0.19705	Closes the remaining point-estimate gap to PIMD8	yes
Nested block selection	Improvement survives selection confined to training folds	same three feature blocks	SMILES	nested strict	0.19931	Confirms the crossing without outer-test block choice	confirmatory
PIMD8 residual correction	Remaining PIMD8 error is learnable	ARROW/PIMD8 + experiment	test-molecule PIMD8	nested	0.18682	Ceiling result; non-deployable	no
OpenFE diagnostics	Simulation convergence and component structure add information	explicit-water archive	ASFE SMILES after distillation	strict	0.20575	Non-improving matched screen	no
MLFF hierarchy	Multiple force-field estimates span missing physics	published MLFF hydration values	SMILES after distillation	strict	0.20109	Non-improving matched screen	no
DES370K/SAPT	Pair interaction components improve water response	water-dimer SAPT subset	SMILES after distillation	strict	0.20993	Pair physics was insufficiently aligned to the endpoint	no
GNNIS force response	Learned explicit-water forces give a useful latent response	public force-matched model	structure	strict	0.20794	Static distilled response did not improve	no
Lambda-aware implicit response	Coupling-dependent potential transfers to hydration	public potential	structure	strict	0.20974	Non-improving	no
Phase-space response	Structural dynamics supply complementary response information	explicit-solvent phase-space averages	SMILES after distillation	strict	0.21894	Non-improving	no
G-NequIP hierarchy	3D solvation representations improve transfer	gas/SMD 3D model	conformer only	strict	0.22276	Non-improving	no
ConfSolv graph latent	Learned latent is richer than summary targets	ConfSolv graph encoder	SMILES	strict	0.21219	Compact response moments transferred better	no
ConfSolv FFN latent	Feed-forward hidden state improves endpoint	ConfSolv encoder	SMILES	strict	0.21548	Non-improving	no
Multi- λ response	A predicted alchemical curve is a better intermediate than one scalar	three-point PIMD2 response	SMILES after distillation	outer-fold strict	0.20136	Response errors remain too large	no
Fidelity hierarchy	Classical, NQE and PIMD heads order the physical target	ARROW classical/PIMD labels	SMILES after distillation	outer-fold strict	0.21474	Sparse aligned labels do not yet transfer precisely	no
Integrated response curve	Predicted $dH/d\lambda$ can be integrated directly	three-point PIMD2 response	SMILES after distillation	outer-fold strict	1.51356	Identifies structure-to-response accuracy as bottleneck	no
Selective simulation	Routing difficult molecules preserves accuracy at lower cost	PIMD8 fallback	PIMD8 for selected inputs	exploratory	varies	Outside final deployment objective; archived for completeness	no

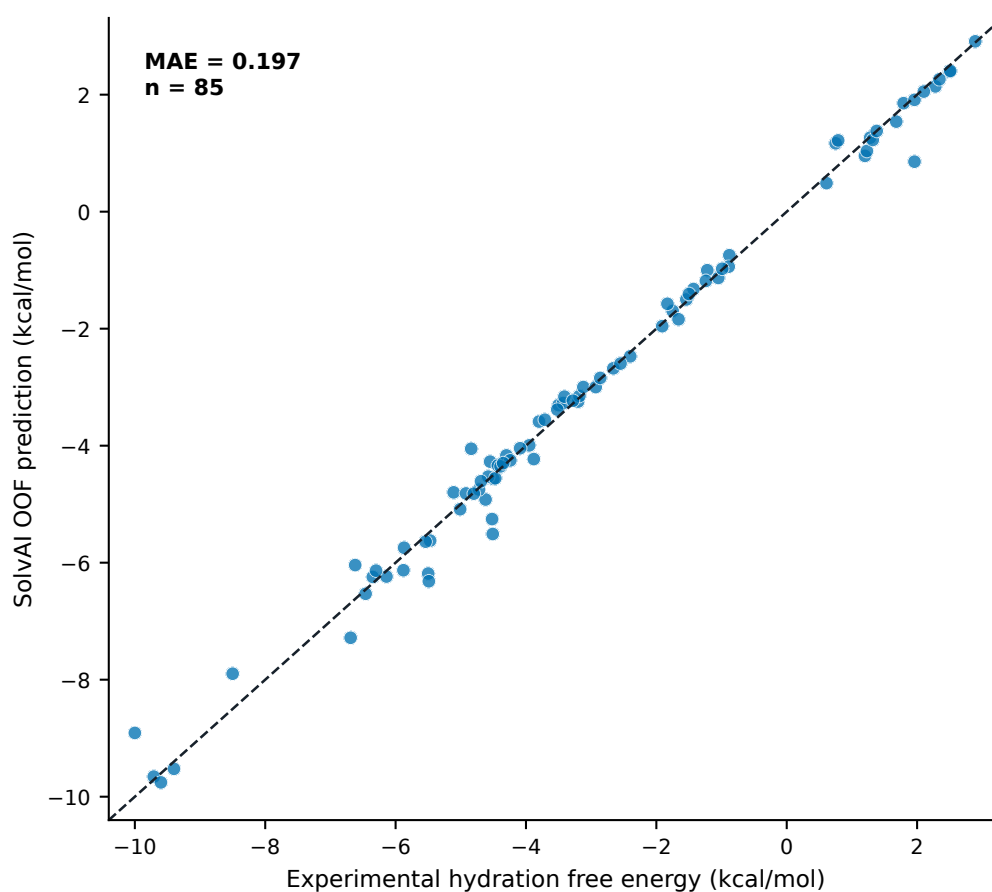


Figure 1: **Extended Data Figure 1 | Experimental and strict OOF predictions.** Each point is produced by a model that did not receive that molecule’s experimental target. The diagonal denotes equality.

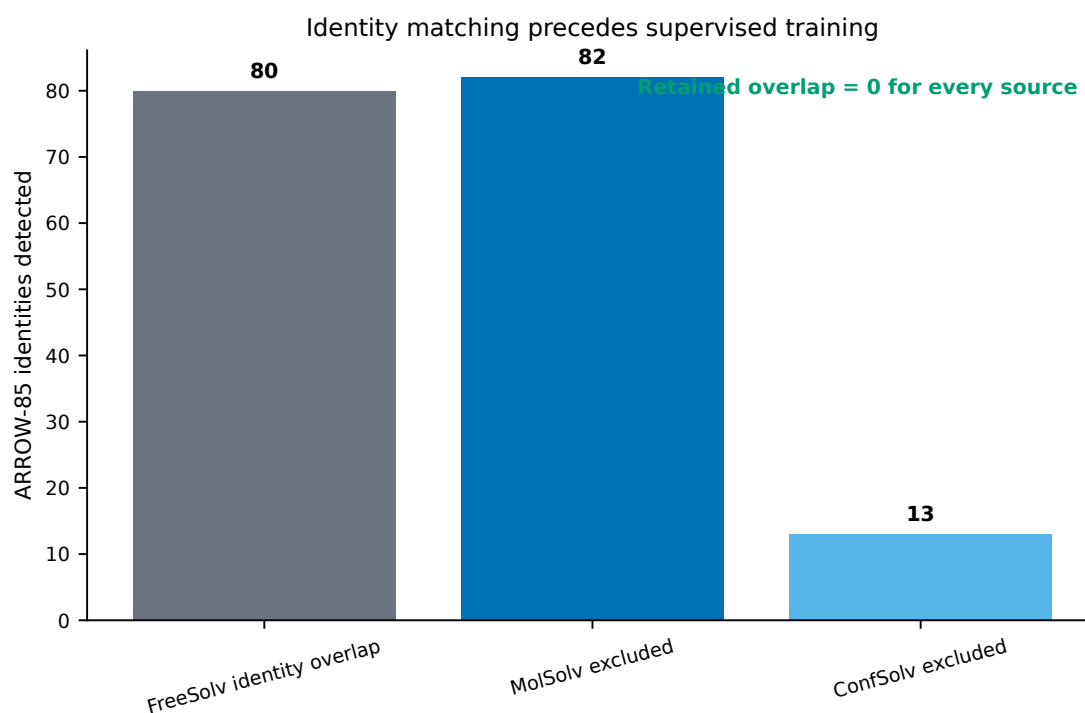


Figure 2: **Extended Data Figure 2 | Benchmark identity analysis and exclusions.** Eighty benchmark connectivities are also present in FreeSolv. Independent global exclusion removed 82 exact MolSolv structures and 13 ConfSolv connectivities; no retained training source overlaps ARROW-85. Counts are identity detections and need not share a provenance.

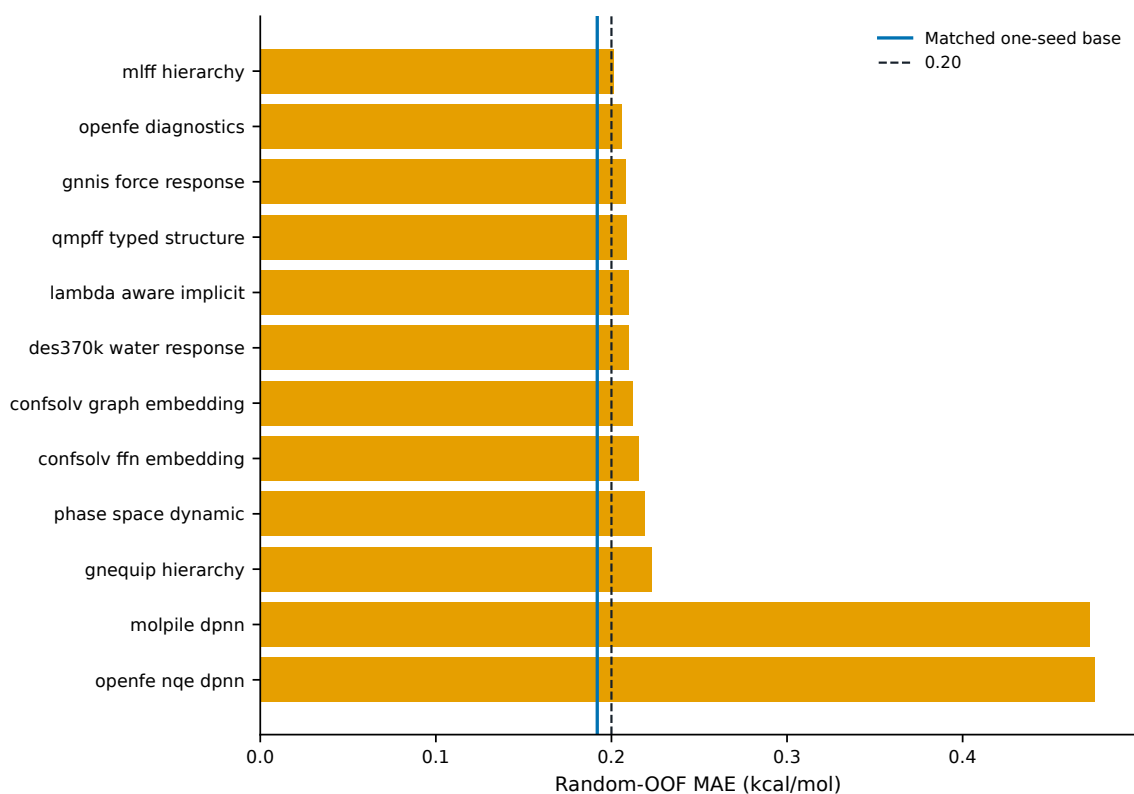


Figure 3: **Extended Data Figure 3 | Alternative physical supervision routes.** Random-OOF MAEs from matched or otherwise frozen screens. These experiments test transfer utility, not the intrinsic quality of each physical method. The matched one-seed base is shown in blue.

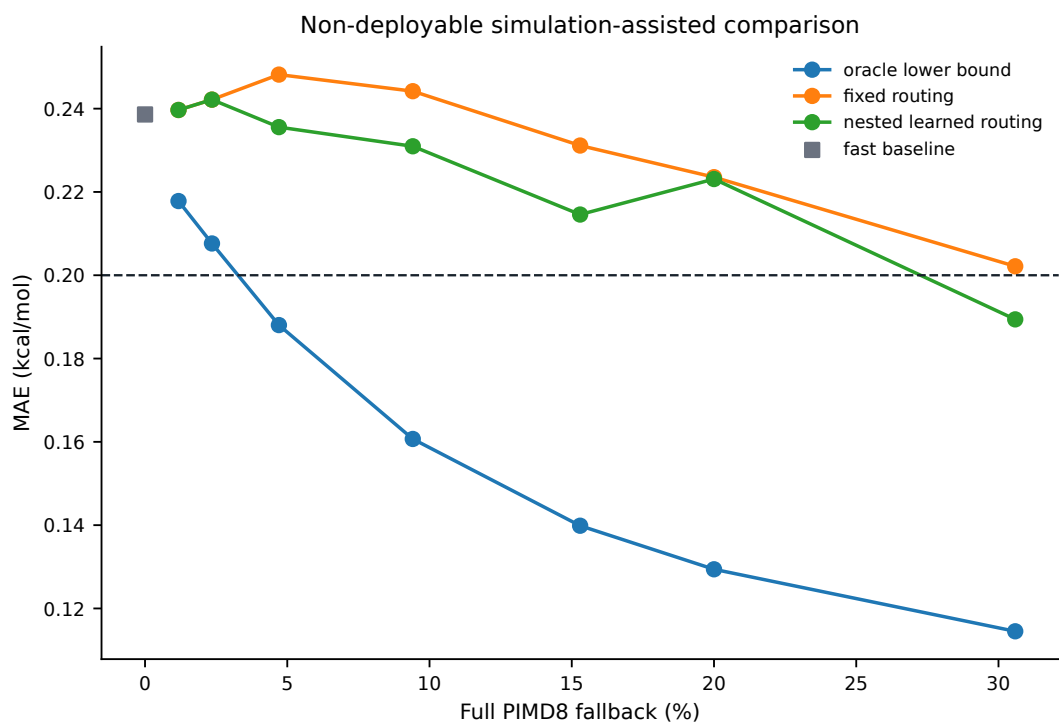


Figure 4: **Extended Data Figure 4 | Non-deployable simulation-assisted comparisons.** Archived selective-PIMD analyses show an oracle lower bound and learned routing under the original random OOF regime. They require PIMD for test molecules and are excluded from SolvAI and all headline claims.

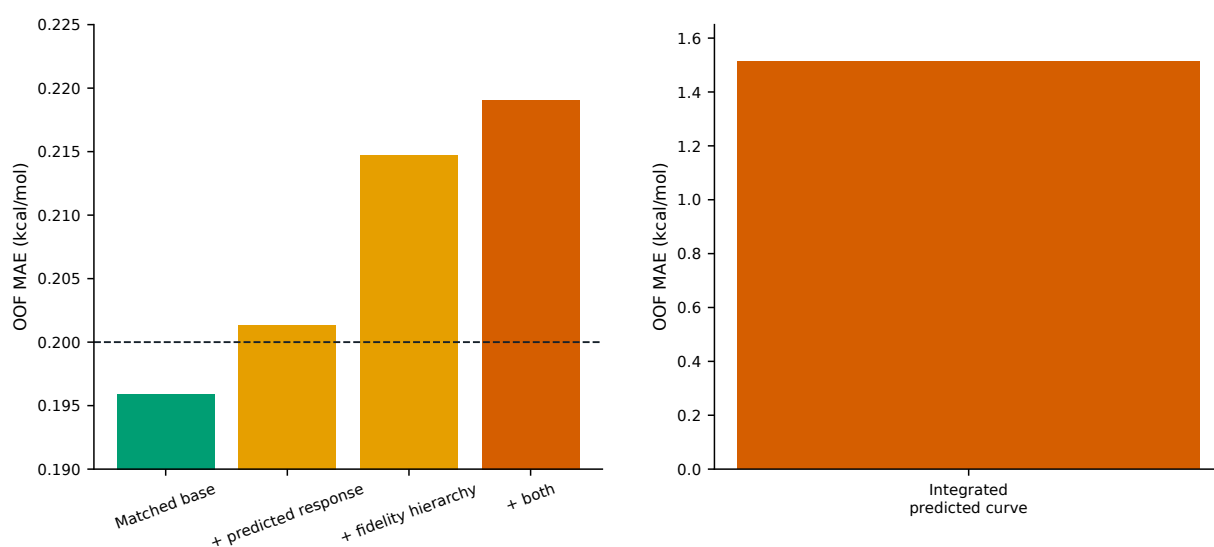


Figure 5: **Extended Data Figure 5 | Multi- λ response distillation.** Predicted PIMD2 response, fidelity hierarchy and their combination all worsened the matched structure-only endpoint. Direct integration of the predicted response was substantially less accurate.

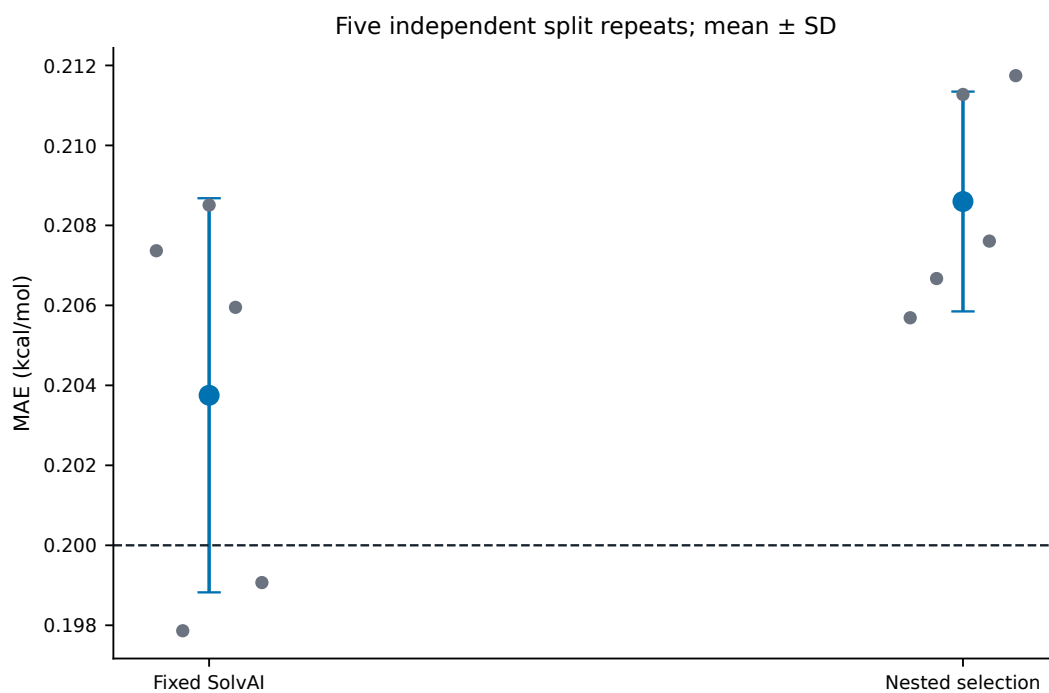


Figure 6: **Extended Data Figure 6 | Independent partition sensitivity.** Points show the five complete repeat MAEs; markers and whiskers show mean and one sample standard deviation.

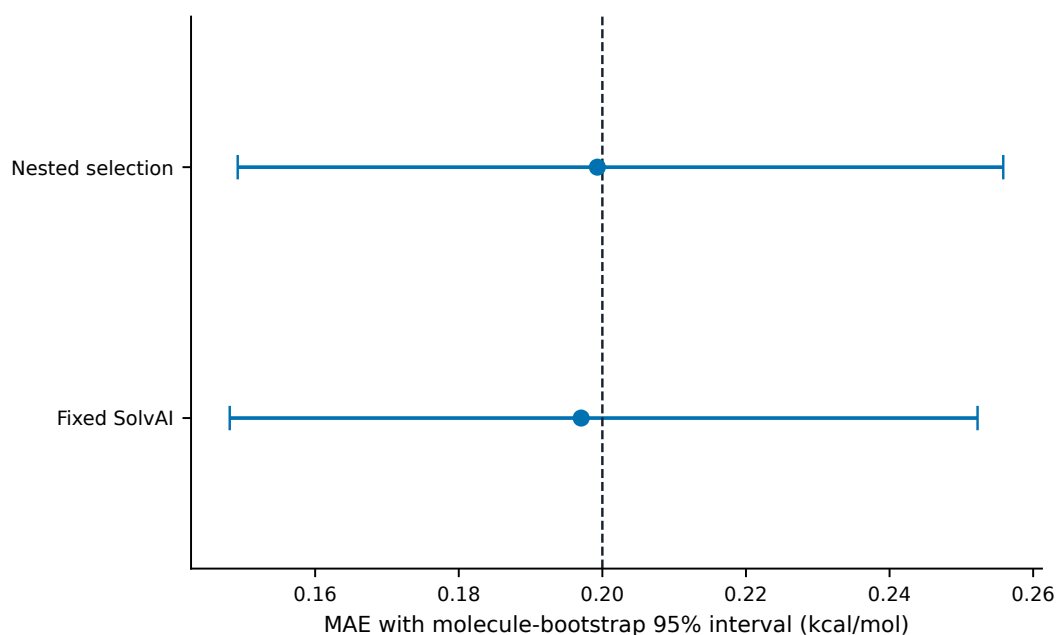


Figure 7: **Extended Data Figure 7 | Molecule-bootstrap uncertainty.** Point estimate and 95% percentile interval from 100,000 resamples. The width reflects the small, heterogeneous reference set.

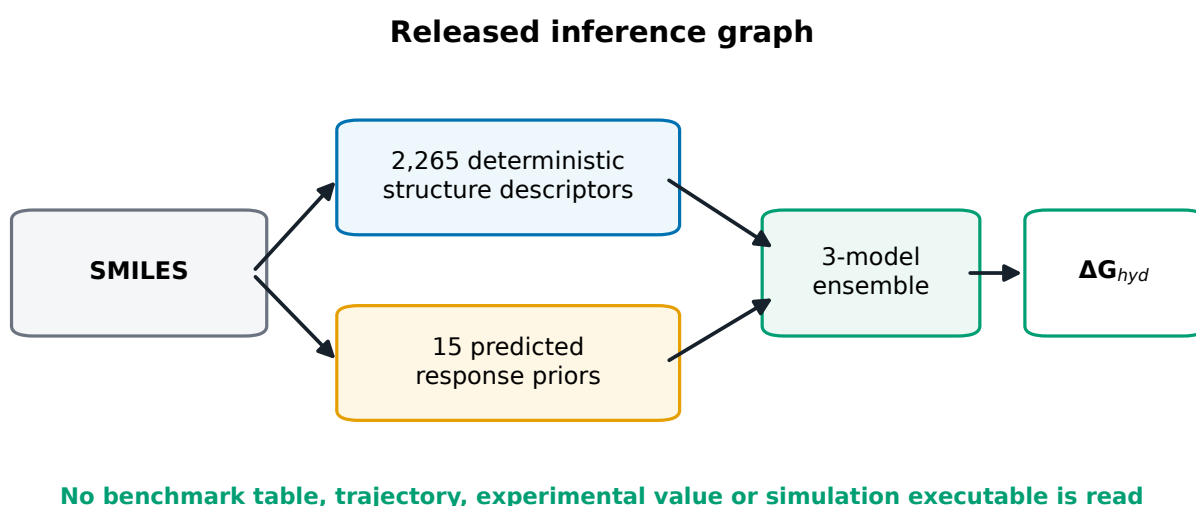


Figure 8: **Extended Data Figure 8 | Released inference graph.** The public artifact receives SMILES, computes deterministic descriptors and structure-derived response priors, and returns an ensemble prediction. Runtime source inspection and schema validation confirm that no benchmark table, trajectory or experimental label is read.

16 **Supplementary Methods**

17 **Teacher-source filtering**

18 For MolSolv, all reference-set connectivity matches were removed before a deterministic size-stratified sample was
19 selected: all eligible molecules with at most ten heavy atoms and one eighth, chosen by SHA-256, with 11–18 heavy
20 atoms. This yielded 350,391 rows and 348,587 unique connectivities. For ConfSolv, H₂O records were grouped
21 by resolved neutral connectivity after removing all benchmark matches. Ensemble and lowest-energy-conformer
22 solution values were converted from kJ mol⁻¹ to kcal mol⁻¹; response summaries were calculated within connectivity.
23 No benchmark target entered either operation.

24 The public endpoint set combined benchmark-disjoint hydration values and retained 1,280 labels using the frozen
25 source-quality mask: membership in the earlier curated table or at least two source measurements. Experimental
26 endpoint rows had unit weight; reference-set outer-training rows had weight three. This weighting and every tree
27 hyperparameter were fixed before confirmatory repeated splits.

28 **Family and scaffold folds**

29 Chemical families were assigned by deterministic structural rules used throughout the campaign. Grouped validation
30 ensures that all members of an assigned family occupy the same held-out partition. Scaffold groups were constructed
31 from the frozen scaffold field; acyclic and small-ring structures were balanced across five groups while preserving
32 identical scaffold membership. These tests are deliberately harder than random OOF and are reported even though
33 neither reaches 0.20 kcal mol⁻¹.

34 **Artifact verification**

35 The model manifest stores SHA-256, byte length and role for each of seven executable artifacts and the model card.
36 Verification checks every hash, then runs ethanol and benzene from SMILES through the public API. Expected
37 predictions are -5.020248 and -0.893576 kcal mol⁻¹, respectively, with absolute tolerance 2×10^{-6} . The test
38 starts no simulator and requires no benchmark file.

39 **Runtime protocol**

40 Cold latency was measured in a fresh Python process. Warm single-molecule latency was measured five times
41 after one warm-up call in the same process. Batch throughput used three runs of 32 SMILES. Timing spans
42 canonicalization, descriptor construction, both D-MPNN processes, all tree teachers and the endpoint ensemble.
43 Median warm latency was 17.522 s and p95 was 17.627 s; median batch time was 17.911 s (0.560 s per molecule).
44 Peak resident memory was 155,408 KiB. The benchmark JSON records exact package, CPU and accelerator
45 information.